📘 SmartSchool Management System

1. Project Overview

The SmartSchool Management System (Principal Panel) is a web-based application designed to manage all school-level academic, administrative, and financial operations in a digital and structured way.

This system enables principals/admin users to manage students, teachers, attendance, exams, results, fees, and reports from a single platform.

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2. Objectives

- **Digitize school administration processes**
- **Reduce manual workload**
- **Maintain accurate real-time data**
- **Improve reporting efficiency**
- **Enable secure data export and backup**

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3. Key Features

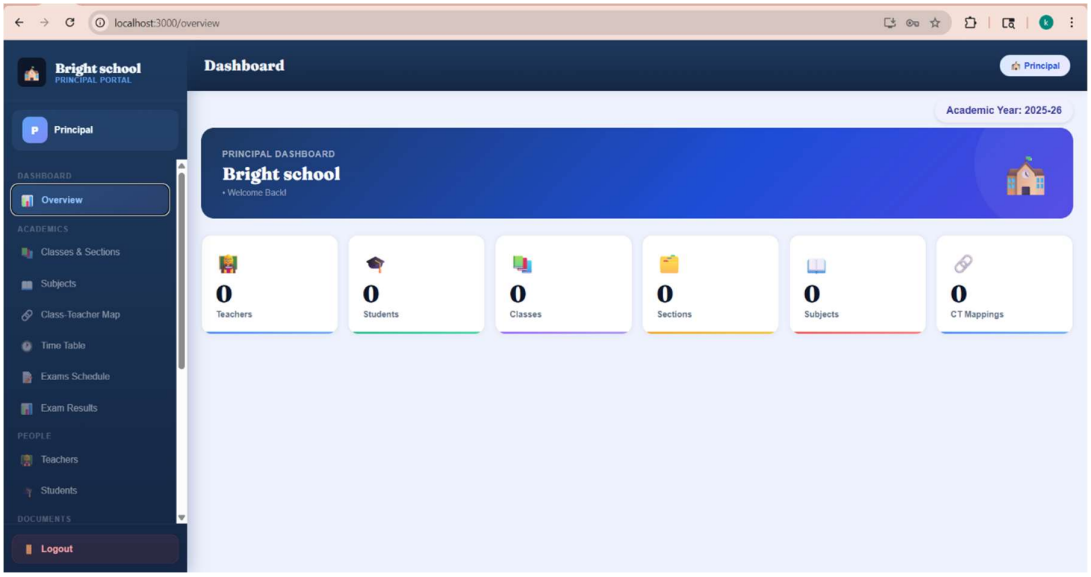
- **Student & Teacher Management**
- **Attendance Tracking System**
- **Exam & Result Management**
- **ID Card Generation**
- **Academic Management (Classes, Subjects, Timetable)**
- **Fee Management**
- **Tally Accounting**
- **Reports Generation**
- **File Management System**
- **Data Export (Excel/CSV)**

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4. System Modules

4.1 Dashboard

Shows overview of students, teachers, attendance, and quick navigation.



Description:

The Dashboard serves as the main landing page for the admin. It provides a summarized view of key school data and system activities.

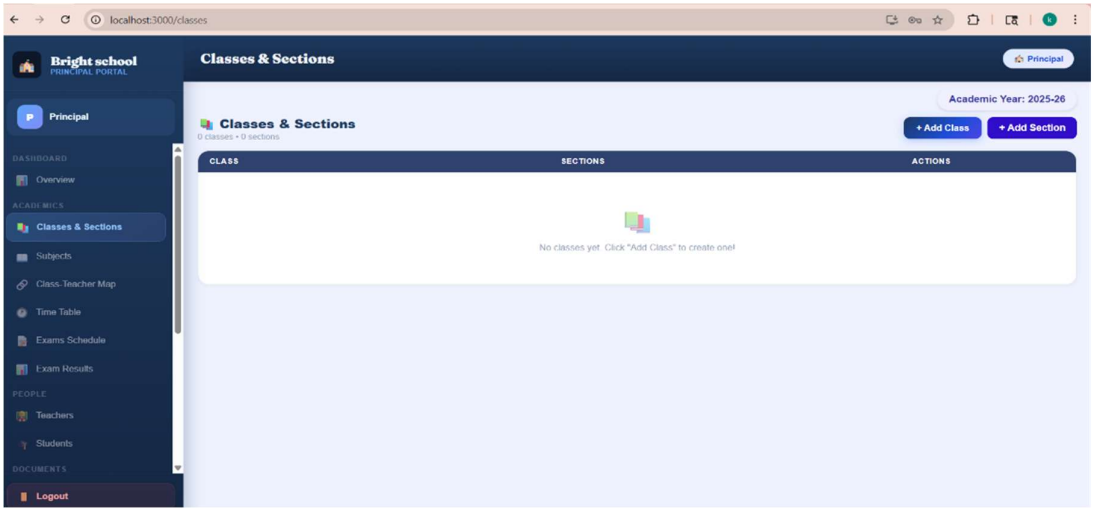
Example:

Upon login, the admin can view total students, total teachers, and overall attendance insights.

Features:

- Overview of key metrics
- Summary cards (students, teachers, attendance)
- Quick access to major modules

4.2 Classes & Sections Management

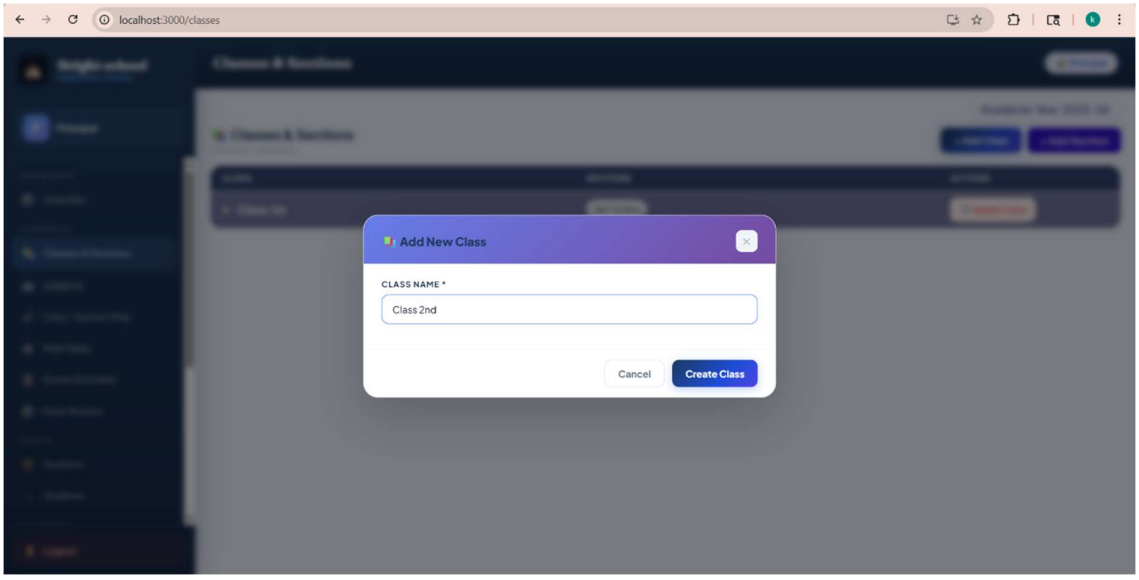


Description:

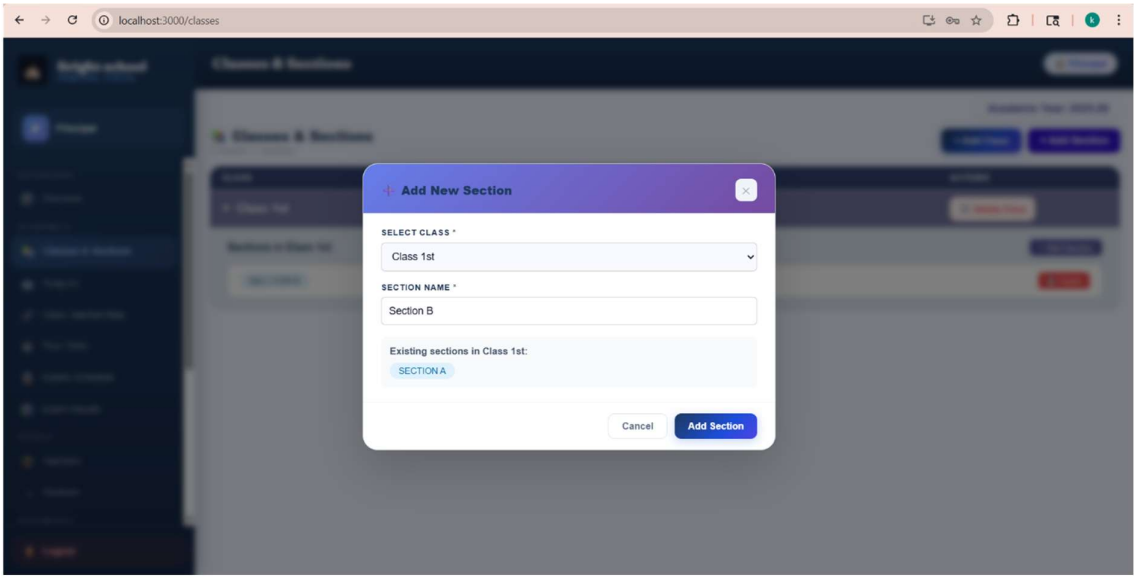
This module allows the admin to create and manage classes and their respective sections in a structured manner.

Example:

The admin can add a new class using the “Add Class” option.
When a class is created, the system automatically assigns a default section (**Section A**) if no section exists.



If the admin wants to add more sections (such as B, C, D, etc.), they can use the **“Add Section”** button available next to the class.



Features:

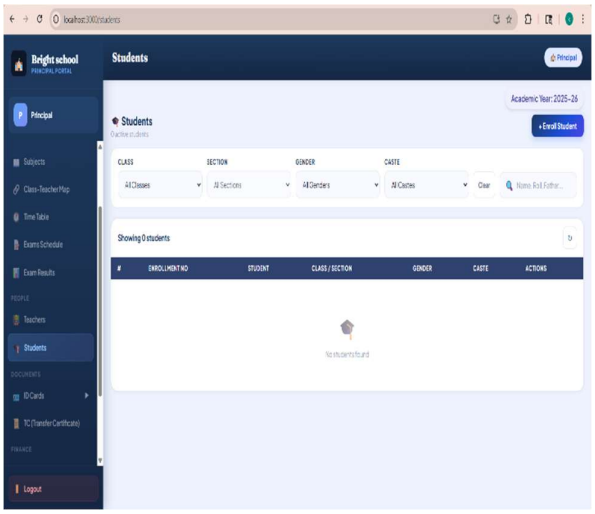
- Add new classes
- Automatic creation of default section (**Section A**)
- Add multiple sections (B, C, D, etc.)
- Section creation via dedicated button
- Prevents duplicate section creation (e.g., Section A cannot be created again if it already exists)
- Structured class-section mapping

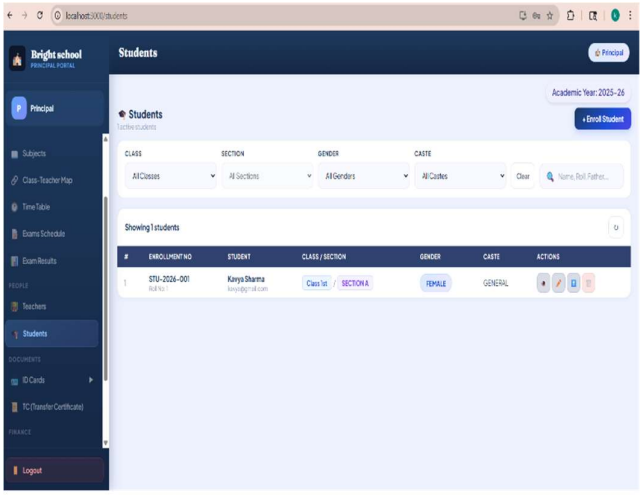
Validation:

- A class must have at least one section
- Duplicate sections are not allowed
- Section names follow a standard format (A, B, C, etc.)

4.3 Student (Add, View, Edit, Delete & Promote)

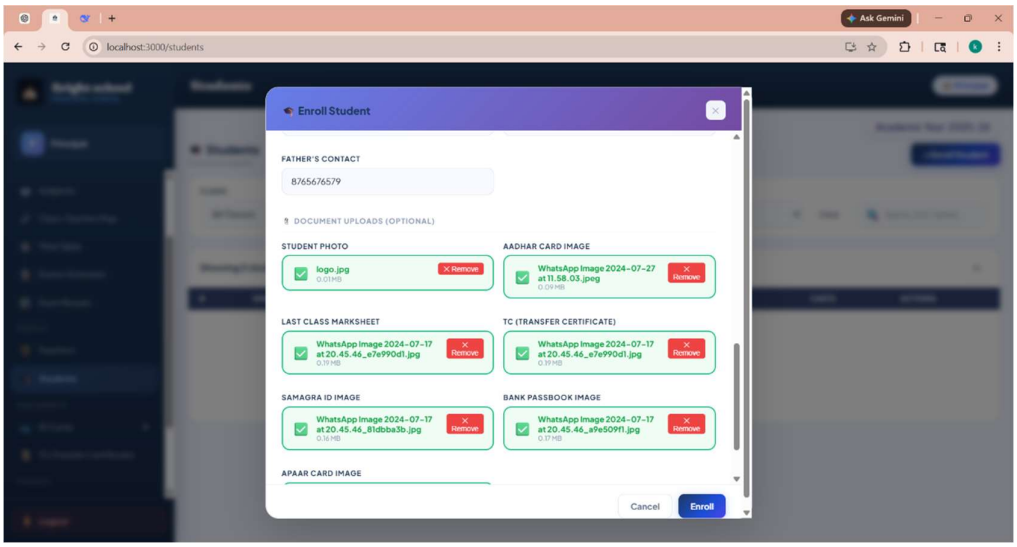
The **Student module** allows the Principal/Admin to manage student records, including adding new students, updating details, promoting students, and performing actions from the student list.





Add Student

- Go to **Students** section
- Click **"+ Enroll Student"**



Fill the student form with:

- Basic details
- Class & Section

- Parent information
- Upload documents (if required)

Click **Save** → Student is successfully added to the system

👉 A unique **Enrollment Number** is automatically generated.

Student List & Actions

After adding, the student appears in the **Student List**, where multiple actions are available under the **Action buttons**:

- **View** → See complete student details
- **Edit** → Update student information (class, section, etc.)
- **Delete** → Remove/deactivate student (with confirmation)
- **Upload** → Add or manage student documents

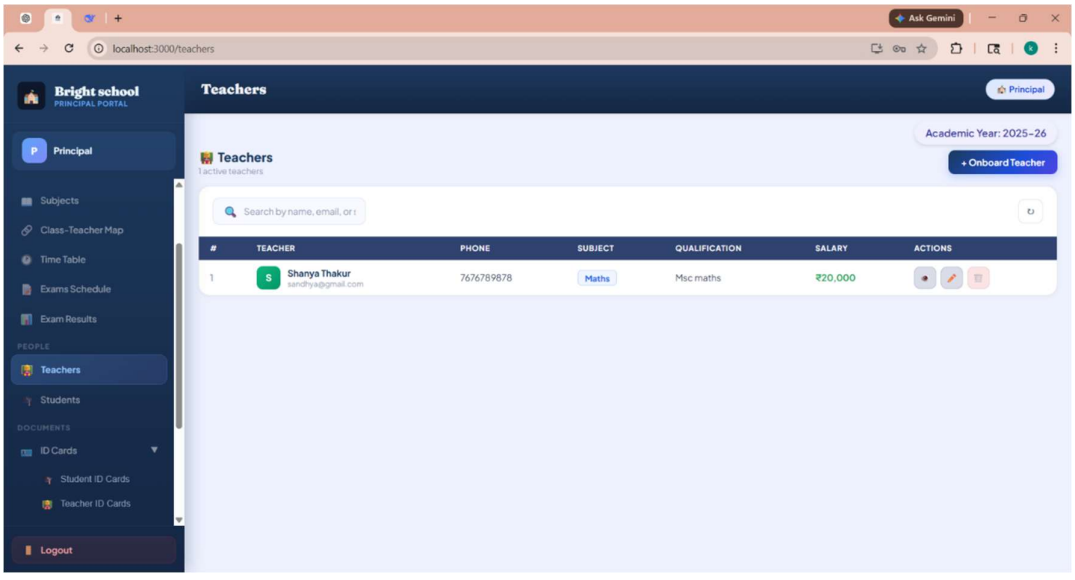
Student Promotion

- Students can be **promoted to the next class/section**
- Enrollment remains the same, only academic details are updated

Key Points

- Students can be added with documents
- All actions are available directly from the list
- Easy management through action buttons
- Data remains structured and updated

4.4 Teacher (Add, View, Edit & Delete)



The **Teacher module** allows the Principal/Admin to manage teacher records, including adding new teachers and performing actions like view, edit, and delete.

Add Teacher

- Go to **Teachers** section
- Click **“Add Teacher”**

Fill the form with:

- Teacher Name
- Subject
- Contact Details
- Upload documents (if required)

Click **Save** → Teacher is added successfully

Teacher List & Actions

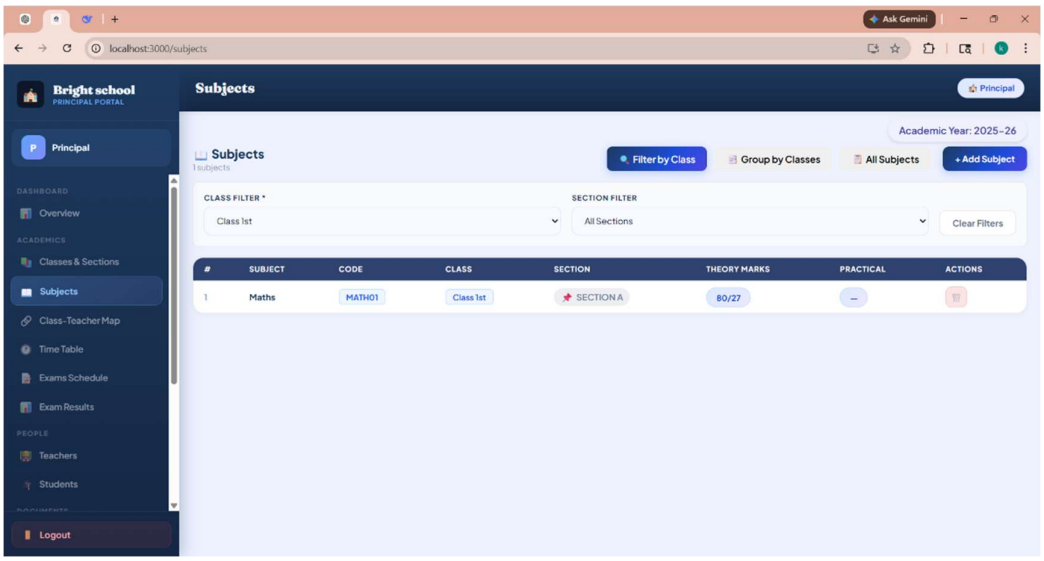
All added teachers appear in the **Teacher List**, where action buttons are available:

- **View** → See complete teacher details
- **Edit** → Update teacher information
- **Delete** → Remove/deactivate teacher (with confirmation)

Key Points

- Easy teacher registration with required details
- Documents can be uploaded and managed
- All actions available directly from the list
- Ensures organized teacher data management

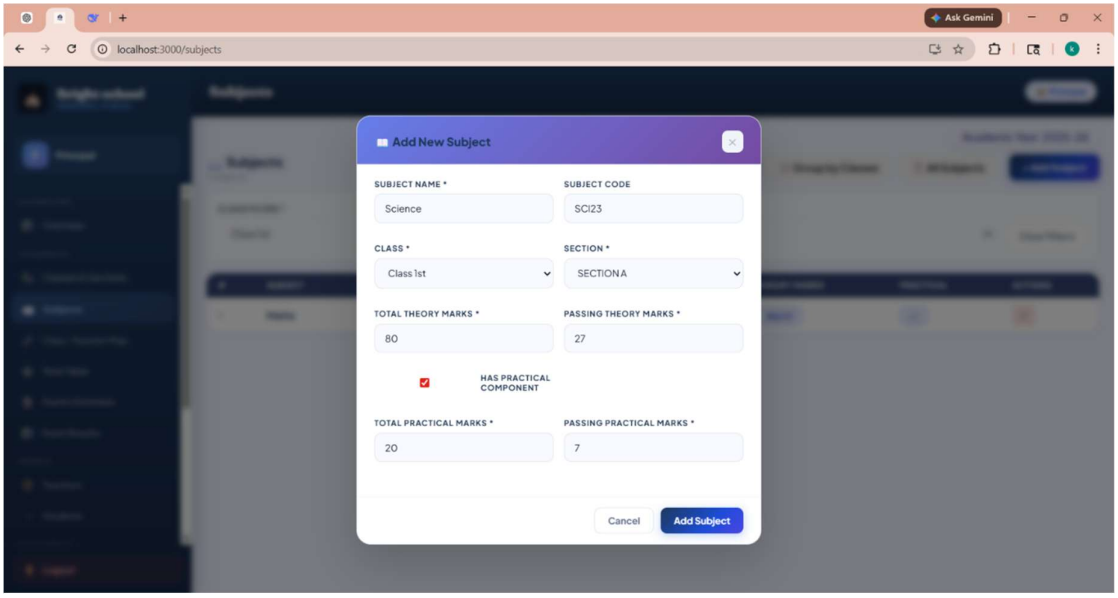
4.5 Subjects



- Yahan sab subjects list me dikhte hain with **class, section, marks details**

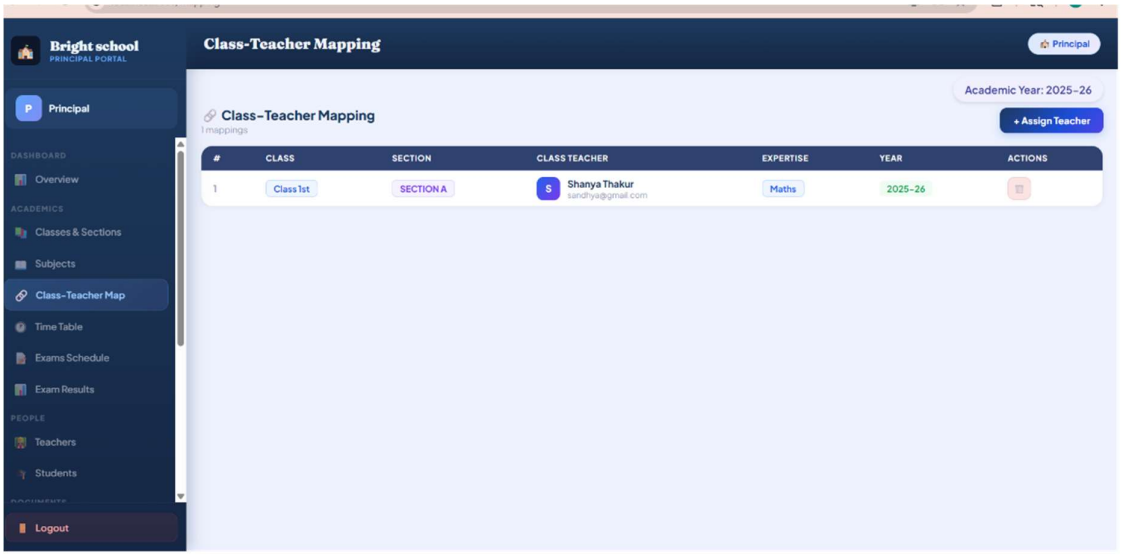
- **Filter by Class & Section** use karke specific subjects dekh sakte hain
- **+ Add Subject** button se naya subject add hota hai

Add Subject Form:



To add a new subject, the **"+ Add Subject"** button is used. In the add subject form, the user enters details like subject name, subject code, class, and section along with theory marks and passing marks. If the subject includes a practical component, practical marks and passing marks can also be added. After filling the form, clicking **Save** will add the subject to the list.

4.6 Class-Teacher Mapping



- This module is used to **assign a teacher to a specific class and section**
- The **"Assign Teacher"** button is used to create mapping

Assign Teacher:

- Select Class
- Select Section
- Select Teacher
- Click Save → Teacher is assigned

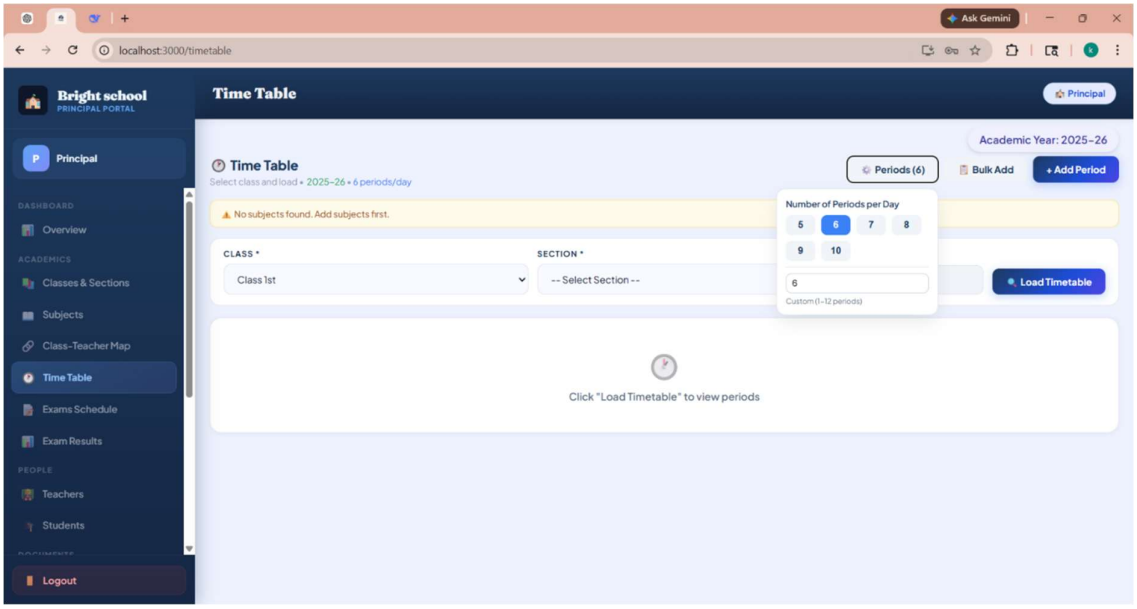
List View:

- All assigned mappings are displayed in a list
- Shows which teacher is assigned to which class/section

Actions:

- Assigned teacher can be removed/deleted if required

4.7 Timetable Management

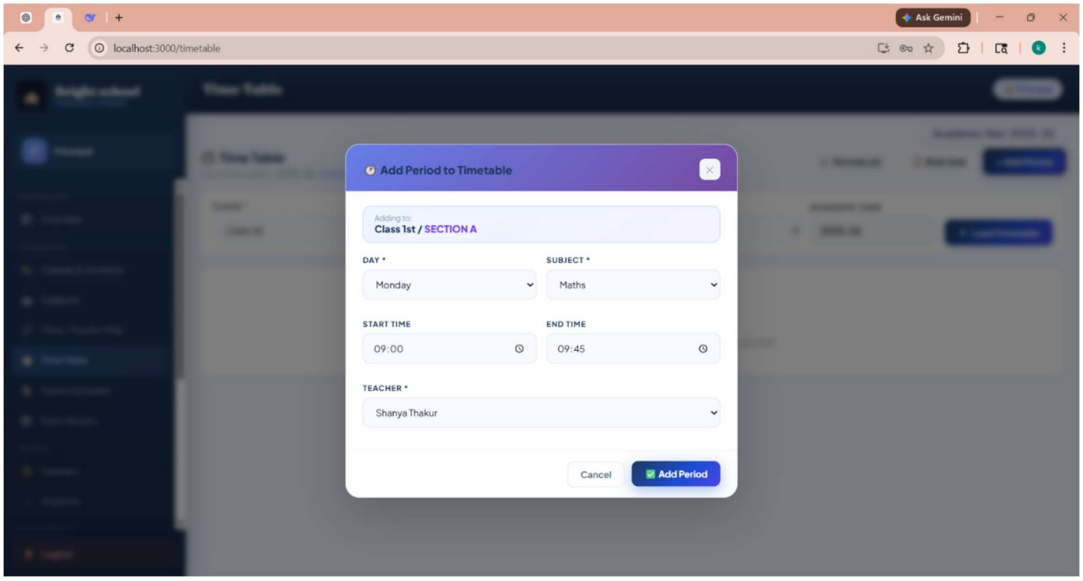


- This module is used to **create and manage class timetables**
- Admin can define periods and assign subjects with timing

Set Periods

- The **"Periods" button** (top) is used to define how many periods are in a school day
- Example: 6 periods, 8 periods, etc.

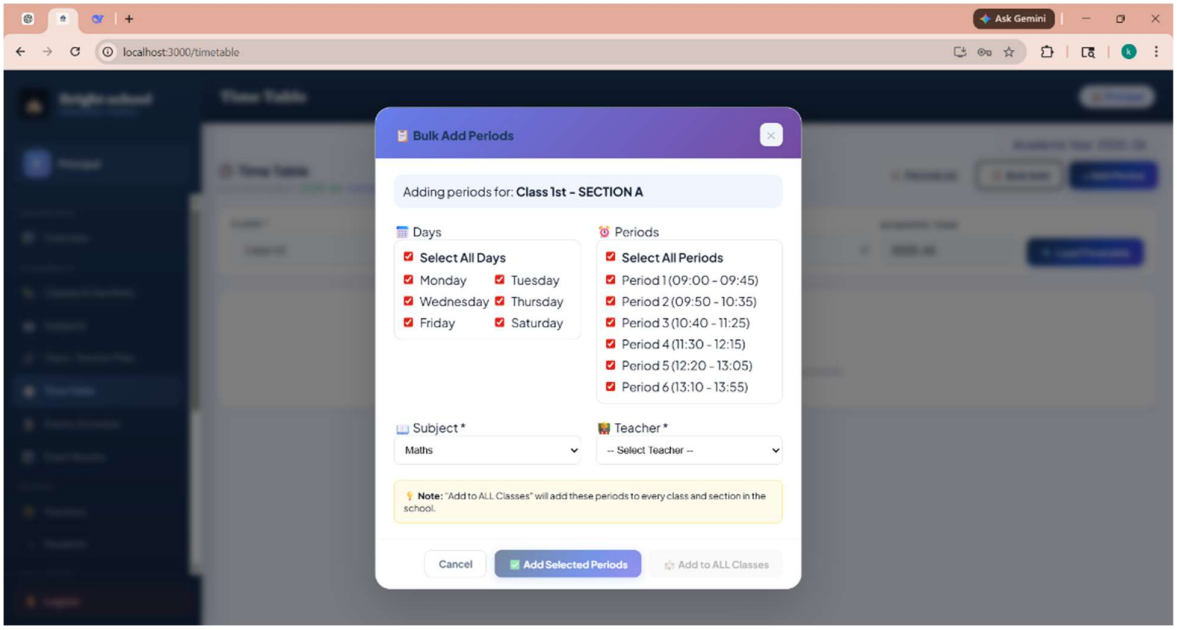
Add Timetable



Single Period

- Add one period at a time by selecting:
    - Class & Section
    - Day (e.g., Monday)
    - Subject
    - Start Time & End Time
    - Teacher
- Click Save → Period is added to timetable

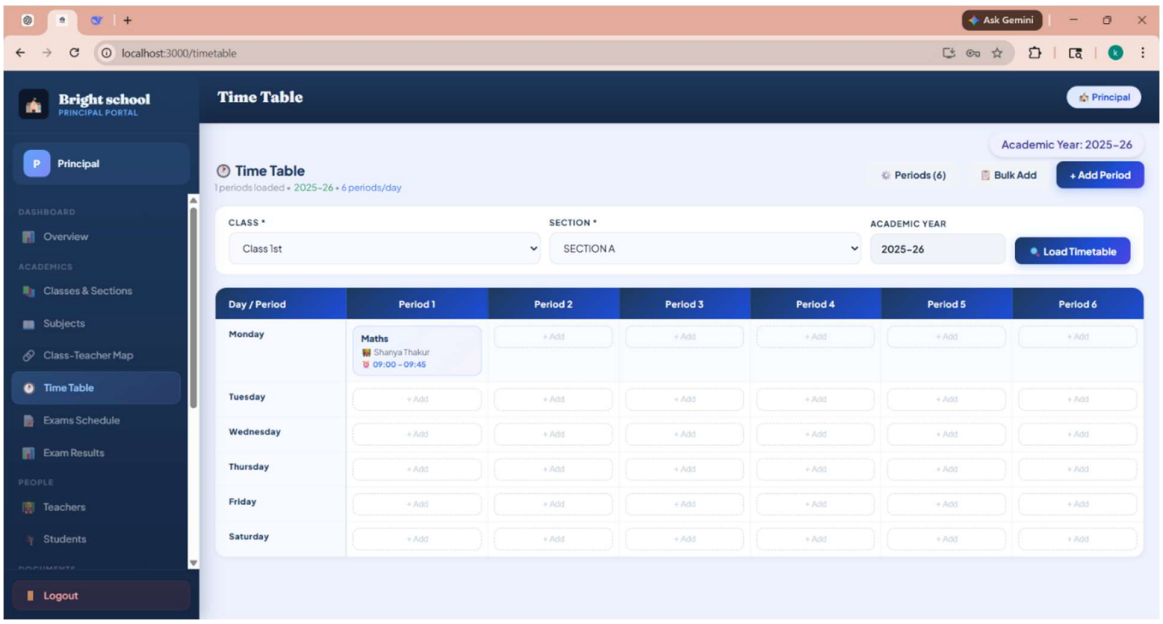
Bulk Add



- Multiple periods can be added together
- Set timings and subjects for all periods at once

- Saves time for full timetable creation

View Timetable

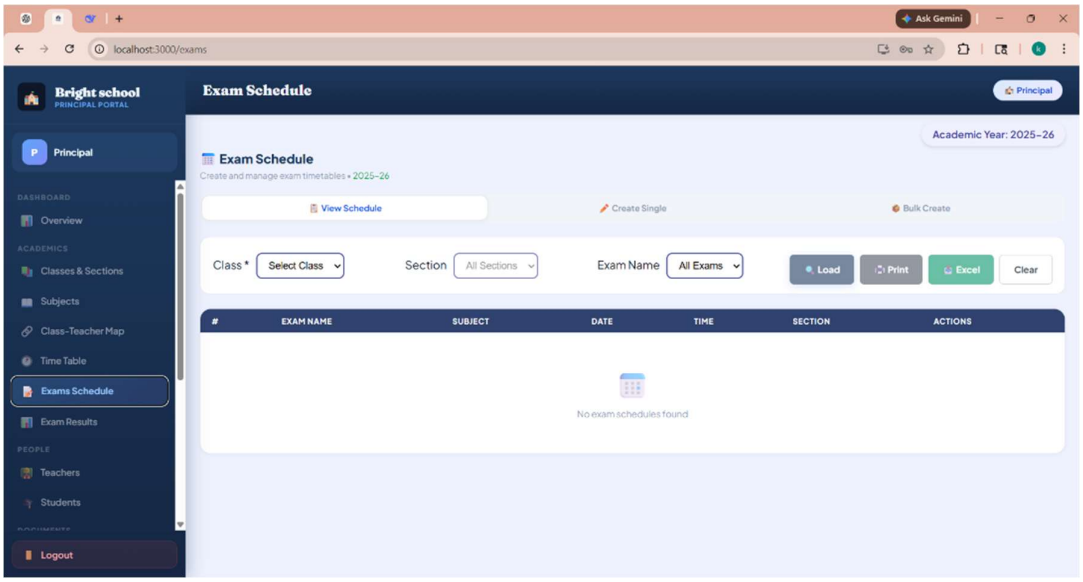


- Use **filters (Class & Section)** to view timetable
- Displays complete schedule for selected class

Key Points

- Periods must be set before adding timetable
- Supports both single and bulk entry
- Easy filtering to view class-wise timetable

4.8 Exam Schedule



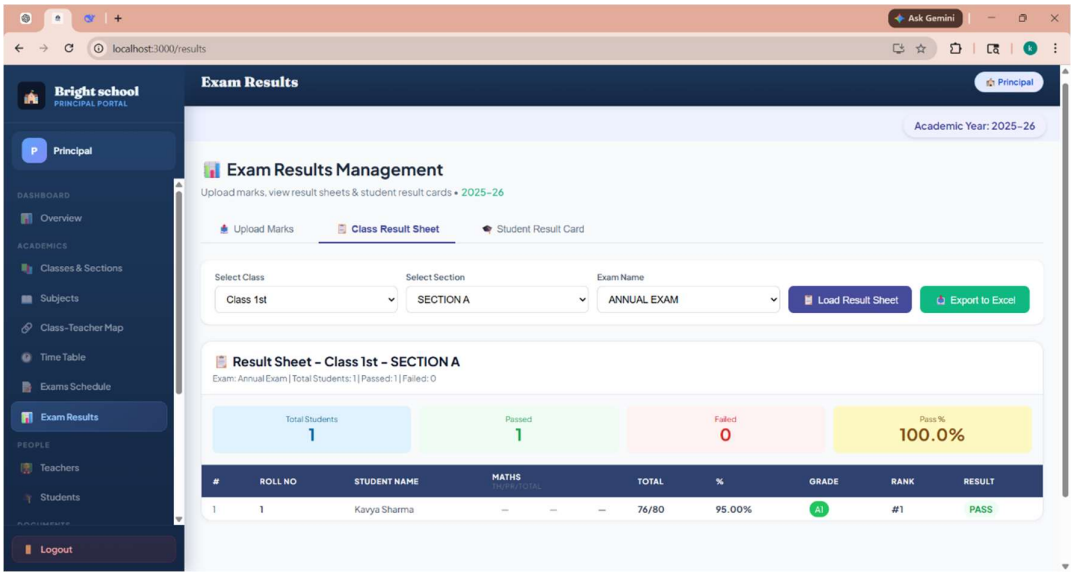
- This module is used to **create and manage exam timetables**

- Exams can be viewed, created, and exported

View Schedule

- Displays exam timetable in table format:
    - Exam Name
    - Subject
    - Date
    - Time
    - Section
- Filters available:
    - Class
    - Section
    - Exam Name
- Buttons:
    - **Load** → Fetch schedule
    - **Print** → Print timetable
    - **Excel** → Download in Excel
    - **Clear** → Reset filters

4.9 Exam Results



- This module is used to **manage and view student exam results**
- Results can be viewed class-wise or student-wise

View Results

- Displays result data such as:
    - Student Name
    - Subject
    - Marks Obtained
    - Total Marks
- Filters available:
    - Class
    - Section
    - Exam Name
- Helps in quickly viewing specific results

Student Result

- View **individual student result card**
- Shows subject-wise marks and overall performance

Bulk Upload

- Upload marks for multiple students at once
- Saves time compared to manual entry

Export

- Results can be downloaded in **Excel format**
- Useful for reports and record keeping

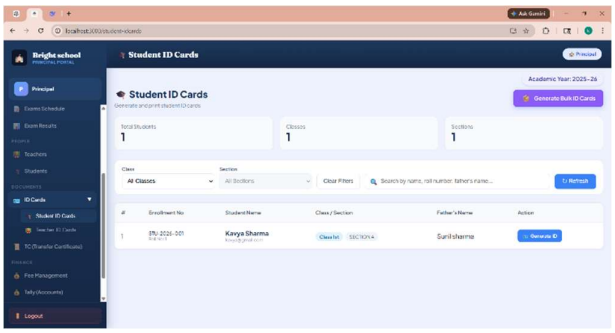
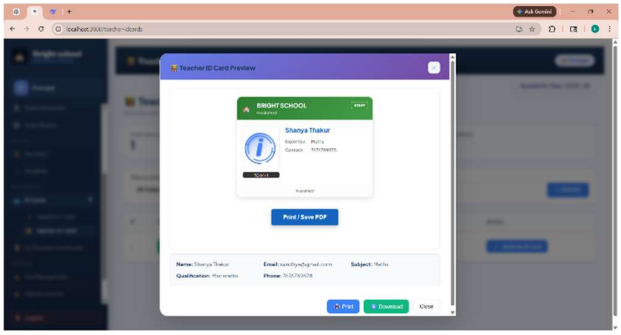
Key Points

- Supports class-wise and student-wise results
- Bulk upload makes data entry faster
- Easy export for reporting

4.10 ID Cards

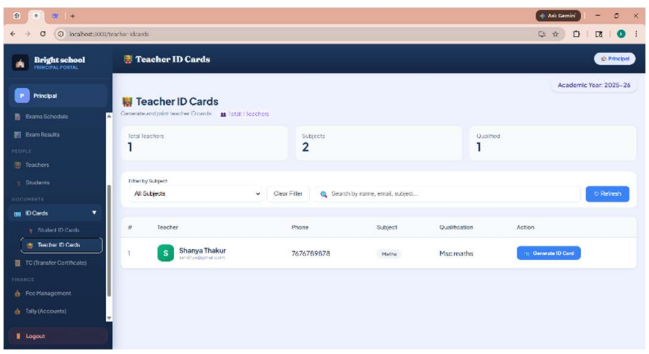
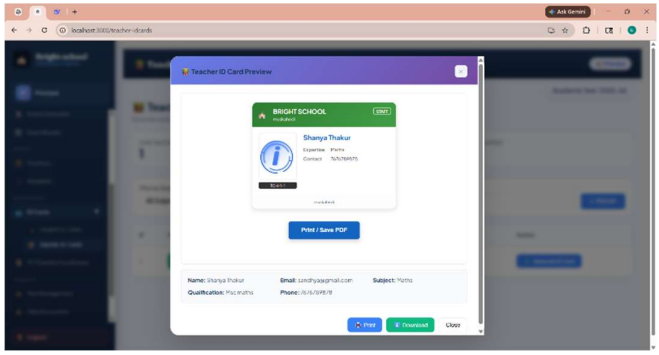
- This module is used to **generate ID cards for students and teachers**
- Supports both **single and bulk generation**

Student ID Cards

- Filter by:
  - Class
  - Section
- View list of students
- Actions:
  - **Generate** → Create ID card for a single student
  - **Bulk Generate** → Create ID cards for all students

Teacher ID Cards

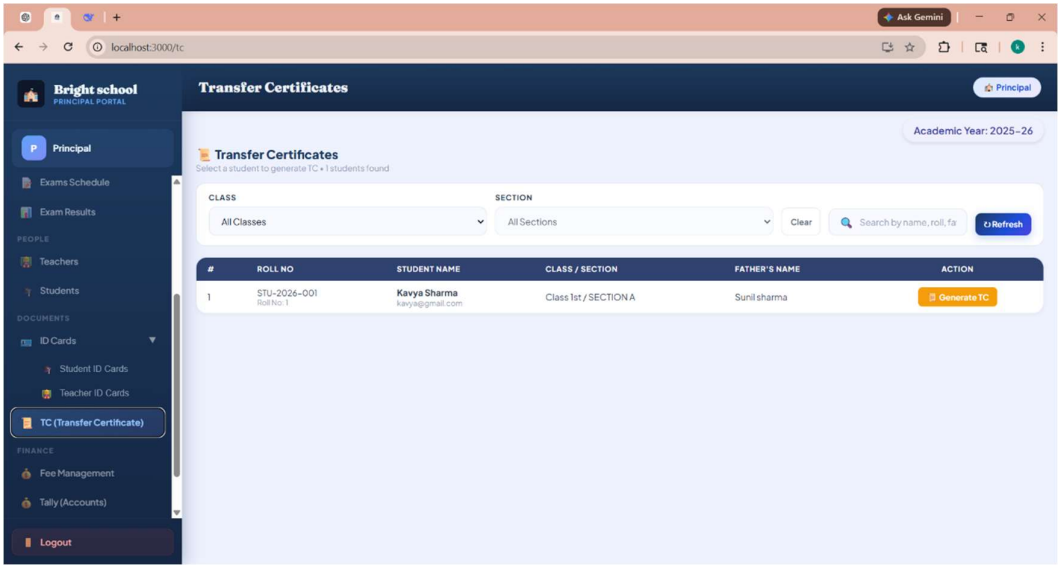



- View list of teachers
- Actions:
  - **Generate** → Create ID card for individual teacher

Key Points

- Quick ID card generation
- Supports bulk download for students
- Useful for official identification

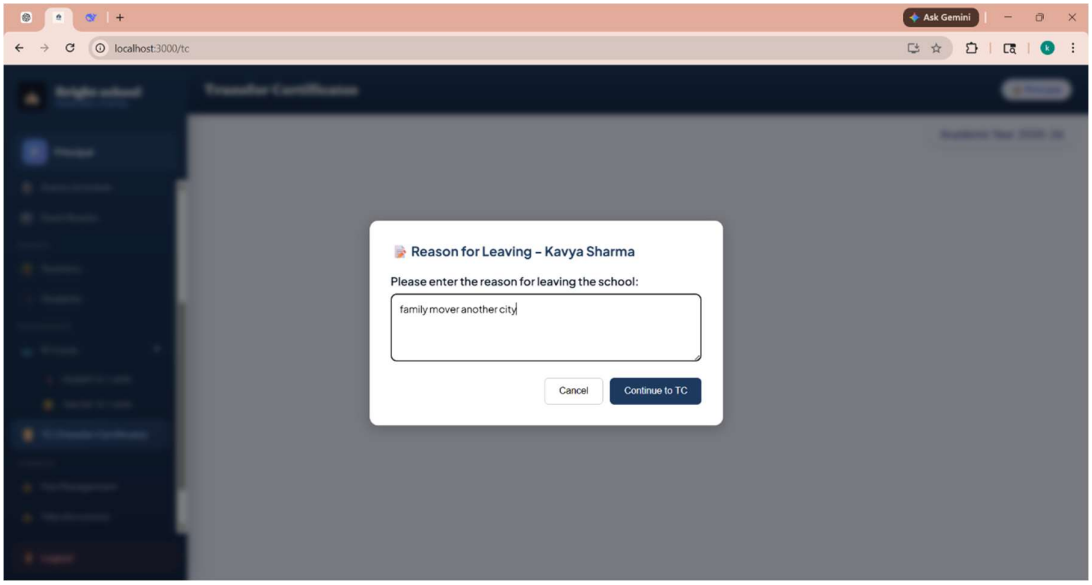
4.11 Transfer Certificate (TC)



- This module is used to **generate Transfer Certificates for students**
- Helps in managing student leaving records officially

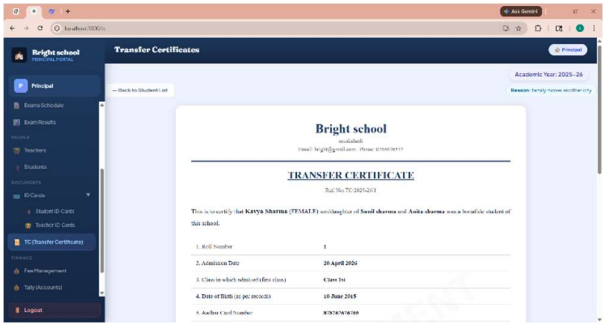
Generate TC

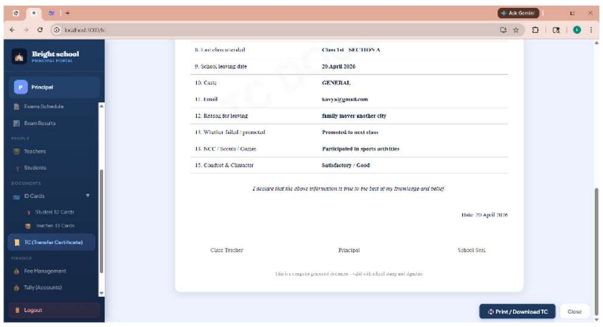
- Select or search student
- System fetches student details automatically
- Generate TC with required information



Features

- Displays student details (name, class, etc.)
- Generates TC in printable format
- Can be downloaded or printed

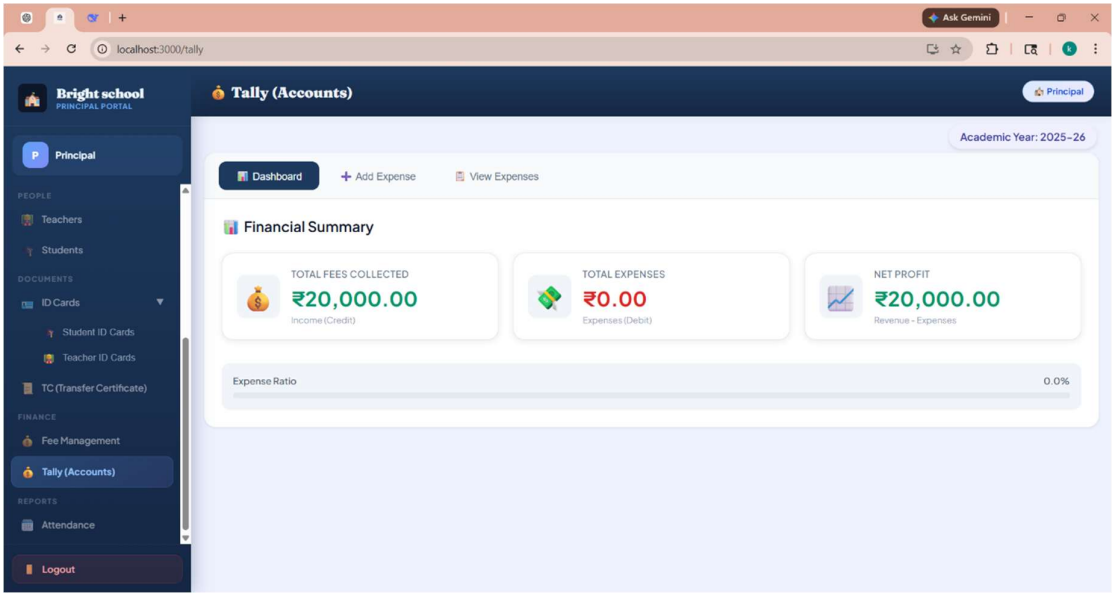




Key Points

- Quick TC generation
- Ensures proper record of student transfer
- Easy print and download option

4.12 Tally (Accounts)

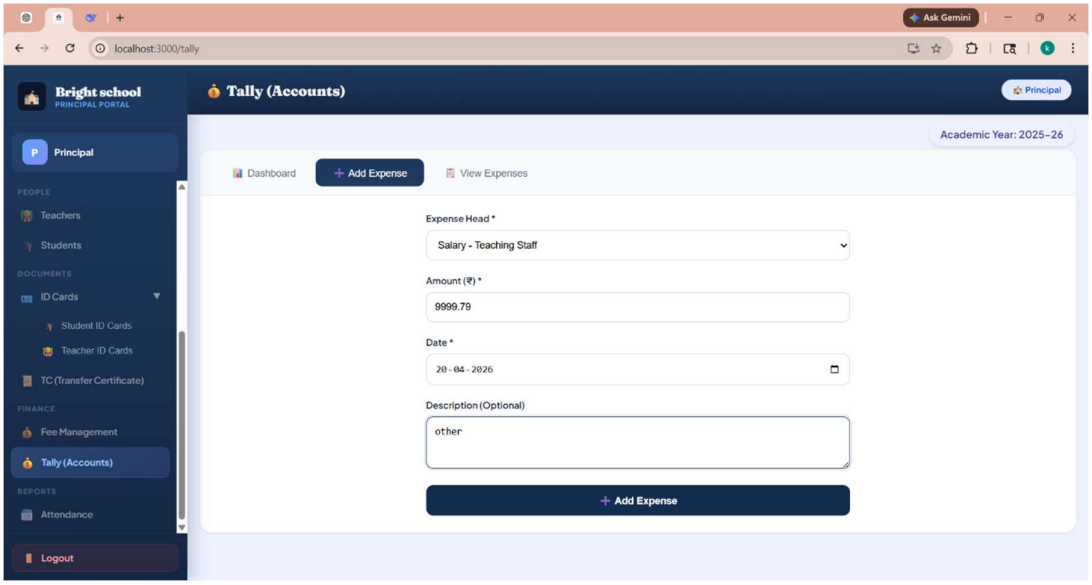


- This module is used to **manage school expenses and view financial summary**
- Helps track income vs expenses for better financial control

Summary View

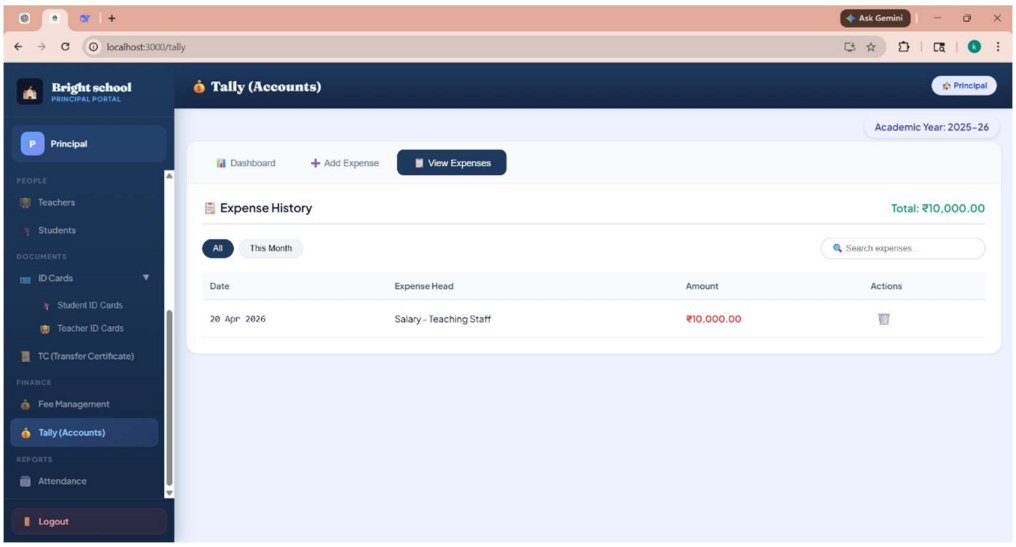
- Shows overall **Profit / Loss**
- Displays total income and total expenses

Manage Expenses



- Add new expense using **Add Expense** button
- Enter details like:
    - Category
    - Amount
    - Date
    - Description
- Saved expenses appear in the list

Actions

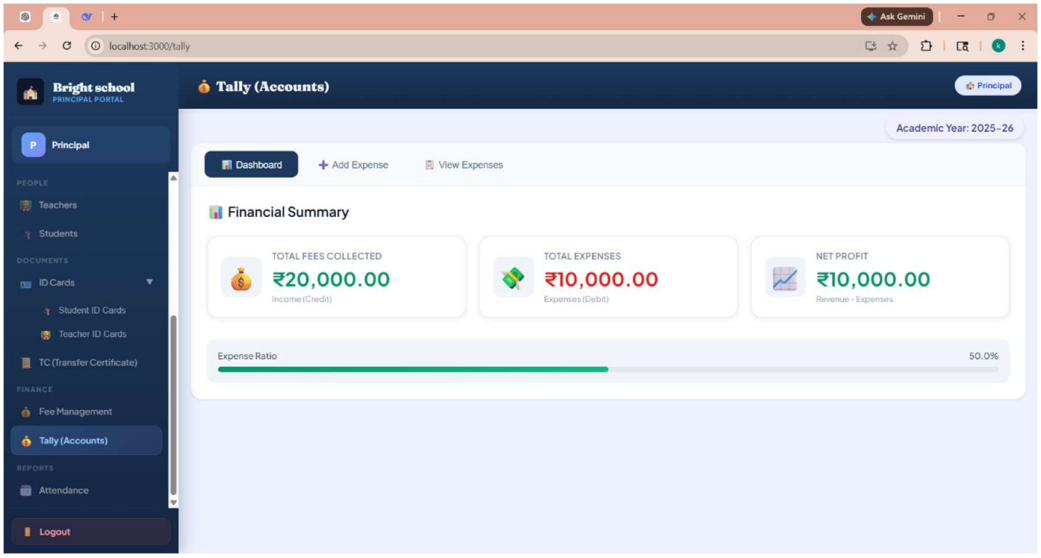


- **Delete** → Remove expense entry if required

Key Points

- Easy expense tracking
- Clear financial overview
- Helps in managing school accounts

4.12 Accounting (Tally)

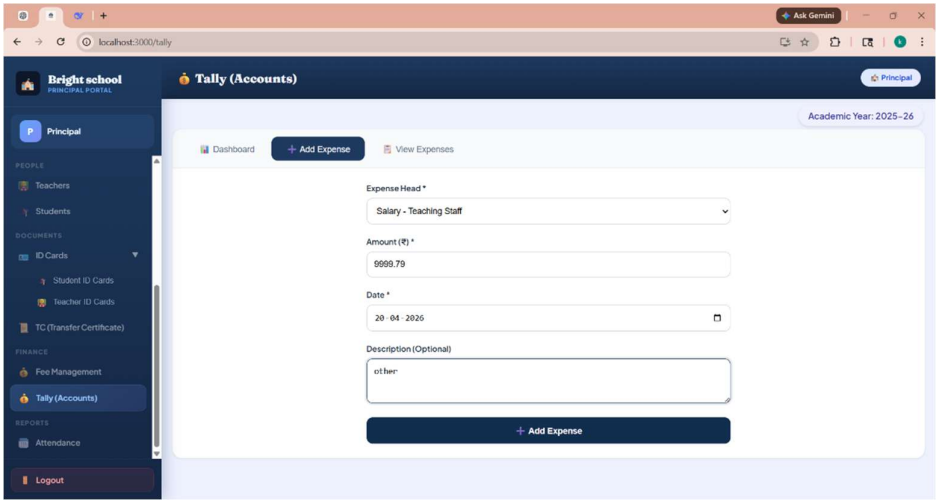


- This module is used to **manage school financial records and track expenses**
- Provides a clear overview of **income and expenses**

Financial Summary

- Displays **Profit / Loss** based on total income and expenses
- Helps in understanding overall financial performance

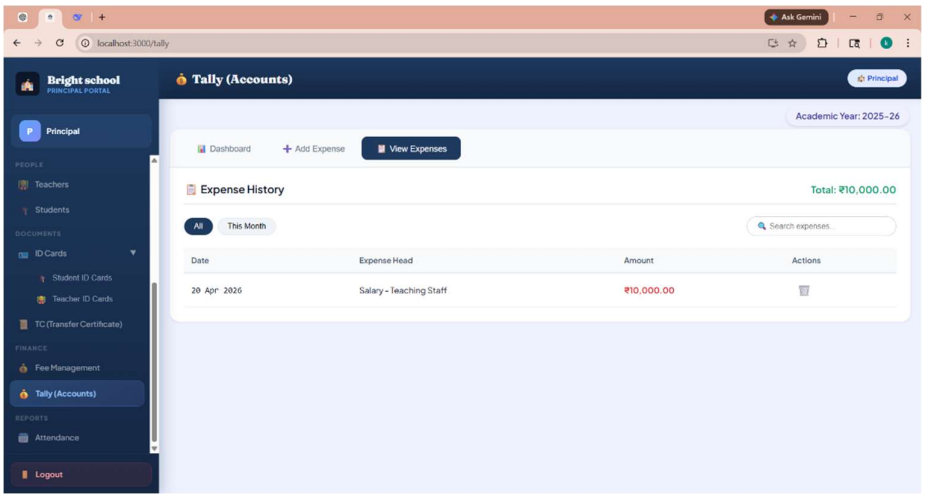
Add Expense



- Use **“Add Expense”** button to record new expenses
- Enter details such as:
    - Category
    - Amount
    - Date
    - Description

- Saved expenses are shown in the list

Actions

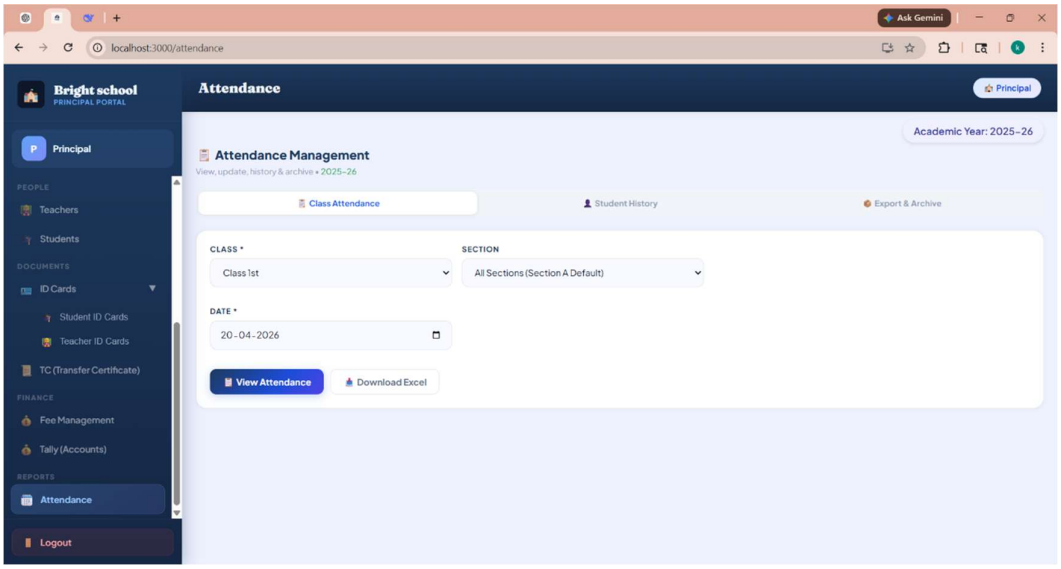


- **Delete** → Remove any expense record if required

Key Points

- Simple and easy expense tracking
- Real-time financial summary
- Helps in maintaining accurate accounting records

4.13 Attendance



- This module is used to **mark and manage student attendance**
- Helps track daily attendance and generate reports

Mark Attendance

- Select:
    - Class

- Section
- Date

- Mark student status:
  - Present
  - Absent
  - Leave
- Click **Save** → Attendance is recorded

View Attendance

- View attendance records using filters:
  - Class
  - Section
  - Date
- Displays student-wise attendance status

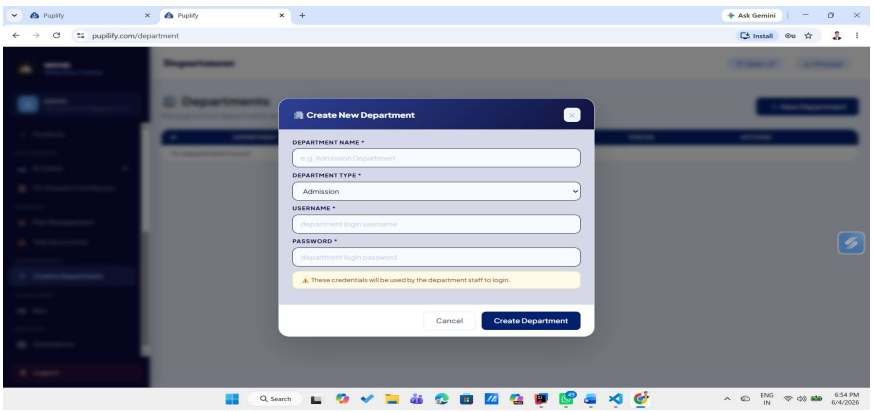
Reports & Export

- Attendance data can be:
  - Viewed in summary
  - Downloaded in **Excel format**

Key Points

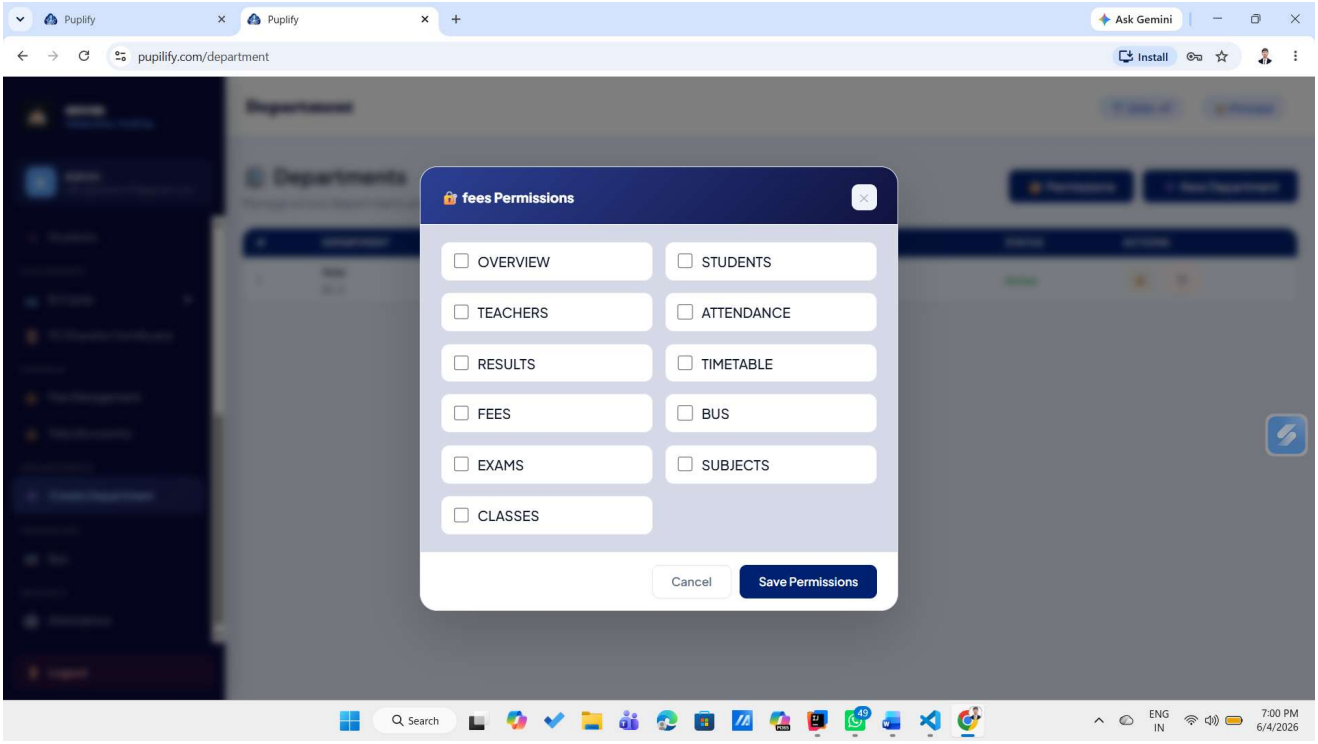
- Easy daily attendance tracking
- Quick filtering and viewing
- Export option for records

4.14 Departments



- First we can create department for give access toh another teacher (like fees, admission, teacher etc)
- It will created by school principle

4.15 Give permission to department



- We can set permission for access the page
- Then it will login or portal with it’s it password

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5. Security & Authentication

- Secure login system (Role-based access)
- Session-based authentication using tokens
- OTP-based password reset system
- Protected API access for all modules

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6. Data Safety Features

- Export data before deletion

- **Confirmation-based delete actions**
- **Secure backup support**
- **Prevent accidental data loss**

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7. System Architecture

The system is designed using a modular architecture:

- **Frontend: React.js (User Interface)**
- **Logic Layer: Controller-based structure**
- **Service Layer: API handling using Axios**
- **Backend: REST APIs (Spring Boot)**
- **Database: MySQL / PostgreSQL**

This ensures scalability, maintainability, and clean separation of logic.

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8. Key Benefits

- **Easy school management in one system**
- **Reduces manual administrative work**
- **Improves accuracy and efficiency**
- **Real-time data availability**
- **Secure and structured data handling**
- **Faster reporting and decision-making**

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9. Conclusion

The SmartSchool Management System (Principal Panel) provides a complete digital solution for managing school operations efficiently.

It helps administrators handle academic, financial, and administrative tasks in a structured and user-friendly way, ensuring smooth school functioning and better data control.